PROJECT & PRODUCT SPECIFICATION

Enterprise GST Input Tax Credit (ITC) Reconciliation, Optimization & Vendor Compliance Platform

Project Codename: ITC-Rec Engine | *Automated ITC Maximize & Statutory Reconciliation Suite*

A single document for two audiences:

Part I — written for the Chartered Accountant, in statutory terms

Part II — written for developers, in plain language

Part III — the commercial pitch & how the software works

Contents

Part I — For the Chartered Accountant

This section is written in the language of statutory compliance and audit practice. It describes what the platform is, the provisions of the CGST Act, 2017 it automates, the day-to-day workflow it replaces, and — importantly — how it differs from the accounting and filing tools already on the market, including the recent wave of AI-based web services.

1. What the Platform Is, In Your Terms

The platform is an automated, continuous, invoice-level reconciliation and audit-verification engine. It bridges the structural disparity between the organisation's internal Purchase Register (PR) / General Ledger (GL) and the data populated on the GSTN portal through suppliers' GSTR-1 / IFF filings — the data that ultimately reflects in the recipient's GSTR-2B and GSTR-2A.

By operating at the invoice and line-item level, the software automates the conditions of Section 16 of the CGST Act, 2017: it optimises Input Tax Credit (ITC) cash flows, eliminates manual spreadsheet dependencies, and maintains a defensible trail for statutory audits and departmental scrutiny.

> **The conditional nature of ITC — the heart of the matter**
>
> A business cannot lawfully avail ITC merely because it holds a valid tax invoice. The credit is conditional: the supplier must declare that specific invoice in their GSTR-1 and discharge the tax, after which it manifests in the recipient's auto-generated GSTR-2B. The gap between “invoice held” and “invoice reflected” is precisely where credit leaks and where exposure arises.

2. The Statutory Problem It Solves

Data asymmetry. Internal purchase registers and the government's GSTR-2B rarely match natively. Keying errors, varied invoice-numbering conventions (INV/26-27/042 vs. 26-27-42 vs. 00042), and cut-off timing differences create thousands of mismatched rows that cannot realistically be resolved by hand on a spreadsheet.

Working-capital leakage. Where a vendor fails to file or under-reports an invoice, the buyer loses the corresponding credit and must discharge that liability again in cash through the monthly GSTR-3B — a direct and avoidable drain on cash flow.

Litigation & penalty exposure. Availing ITC on an invoice the supplier never reported risks contravention of Section 16, inviting departmental scrutiny (Form ASMT-10), penalties, and mandatory interest under Section 50.

3. Core Operational Modules — From a CA Workflow Perspective

A. Dynamic Two-Way Reconciliation (PR vs. GSTR-2B)

In place of static v-lookups that fail on formatting variance, the engine runs a multi-pass strategy and classifies every inward supply into a definitive bucket:

- **Eligible & matched credit —** GSTIN, invoice number, date and tax heads (CGST / SGST / IGST / Cess) all agree. These are cleared for the GSTR-3B draft automatically.

- **Mismatched records (value discrepancies) —** invoice numbers agree but tax figures differ, often due to rounding or partial vendor reporting. The engine computes the net impact to prevent both excess claims and avoidable cash blockage.
- **Timing differences (unreported by vendor) —** invoices booked in the GL but absent from the current GSTR-2B are categorised as "ITC Deferred / Pipeline Credit" to keep the claim within the discipline of Rule 36(4).

B. Intelligent Invoice-Number Normalisation & Fuzzy Logic

A perennial source of friction is a vendor recording INV/26-27/042 where the client's ERP holds 26-27-42 or 00042. An algorithmic text-matching protocol strips non-alphanumeric delimiters and leading zeros to surface structural matches, then flags them to the reviewer as "Probable Matches" with a confidence score — so legitimate credit is not wrongly written off as missing.

C. Compliance Tracking under Section 16(4) & Rule 37

- **Section 16(4) limitation sentinel —** tracks the statutory outer limit for availing credit for a financial year (the earlier of 30 November following the end of that year, or the filing of the relevant Annual Return) and flags ageing unmatched invoices so no eligible ITC lapses permanently.
- **Rule 37 / 180-day reversal monitor —** tracks ageing of unpaid vendor payables against invoice dates; where payment crosses 180 days it prompts reversal of the corresponding ITC with applicable interest, and flags the line for re-claim once payment is cleared.

D. Automated Vendor Communication & Recourse Management

When a mismatch or missing invoice is localised, the software maps it to the responsible vendor and dispatches a system-generated GST Compliance Discrepancy Notice stating the affected invoice details, the tax impact on the enterprise, and a request for amendment in the supplier's subsequent GSTR-1. A vendor dashboard lets suppliers upload missing details directly, reducing the CA's role from data-chaser to overseer.

4. Statutory Audit-Trail Assurance

Every action — a manual force-match of two disparate invoices, a deferred-credit categorisation, or an automated vendor communication — is captured in a tamper-evident audit log. When the department initiates scrutiny under Form ASMT-10, the platform generates an Invoice-Level Reconciliation Statement that explains every variance between the books and the portal, satisfying both internal financial controls and external statutory auditors. The same continuous monthly data converges into an annual reconciliation ledger, making GSTR-9 / 9C preparation materially less stressful.

5. How This Differs From the Tools You Already Use

The market splits into three legacy layers and a newer fourth. The platform is positioned deliberately against each.

Category	What it does well	Where it falls short
Accounting giants (TallyPrime, Zoho Books)	Primary books of account; basic GSTR-2B JSON import.	Reconciliation is purely deterministic (exact match). A slightly different invoice string fails

Category	What it does well	Where it falls short
		to match, forcing manual export to spreadsheets.
Compliance portals (ClearTax, GSTHero, Optotax)	Strong at filing returns (GSTR-1, 3B, 9).	Reconciliation is reactive — run once near the deadline. They hand you a long Excel of mismatches but leave vendor follow-up and dispute management to your team.
Enterprise core (SAP, Oracle)	Heavy-duty transactional engines for large corporates.	Rigid. Custom GST reconciliation logic takes months and high cost to build, and adapts slowly to frequent GSTN portal changes.
New AI / GenAI web services	OCR of invoices, chat-style Q&A and summaries over tax data.	General-purpose probabilistic models are hard to make auditable; a credit allowed by an opaque “best guess” is difficult to defend in scrutiny.

Our point of difference

We treat compliance as a recovery stream, not a cost centre. The engine leads with deterministic, explainable matching and uses financial-tolerance and fuzzy/proximity logic only as a confidence-scored, reviewer-visible assist — never as an unaccountable “black box.” Every match, force-match and adjustment is logged immutably, so each rupee of credit is traceable. Unlike filing-first portals, it closes the loop: it does not merely report a mismatch, it pursues the vendor automatically until the credit is recovered, while watching the Section 16(4) clock so nothing lapses.

Against AI-only web services specifically: those tools are valuable at the edges (reading a scanned invoice, answering a question), but a statutory filing demands determinism and a defensible trail. Here, AI assists; deterministic rules and a tamper-evident log decide. That combination is what makes the output safe to file and safe to defend.

Part II — Developer Brief (Plain Language)

> *If you have never touched Indian tax law, read this part first. It explains the same product in everyday words, defines the jargon, and then describes the system the way you would actually build it.*

6. The One-Paragraph Version

In India, a business can only subtract the tax it paid on its purchases from the tax it owes the government — but only if the company that sold to it also reported that exact sale to the government. Our software pulls in both sides: what the company says it bought (from its accounting system) and what the government says was actually reported (from the official portal). It then matches them invoice by invoice to find what's missing, what's wrong, and what's about to expire. When it finds a problem caused by a supplier, it automatically contacts that supplier and asks them to fix it — and it keeps a complete, tamper-proof record of everything so the company can prove its numbers during an audit.

7. Key Terms, Decoded

Term	What it actually means for the code
ITC (Input Tax Credit)	The money the buyer is allowed to deduct. This is the asset we are trying to protect and maximise.
Purchase Register / GL	The buyer's own list of purchases, pulled from their ERP/accounting tool. "What we think we bought."
GSTR-1	The seller's filing that lists their sales. The source of truth on the supplier side.
GSTR-2B / 2A	The government's auto-generated statement of purchases reported against the buyer. "What was actually reported."
GSTR-3B	The monthly return where the buyer finally claims credit and pays tax. Our "safe to claim" output feeds this.
GSTR-9 / 9C	The annual return and reconciliation. We roll monthly data up into this.
GSTIN	The 15-character tax ID of a business. A primary key for matching.
Section 16(4) / Rule 37	Deadlines. 16(4) = last date to claim; Rule 37 = reverse credit if the supplier isn't paid within 180 days. Both are timers we monitor.
ASMT-10	A scrutiny notice from the tax department. Our audit trail is what answers it.

8. How It Works — Functional Architecture

The system runs as four automated layers. Think of it as: get the data, match the data, chase the problems, show the results.

Layer A — Enterprise Data Ingestion

- Connects via secure REST APIs to ERPs (SAP, Oracle, Microsoft Dynamics) and cloud accounting tools (Tally, Zoho) to pull the internal Purchase Register.

- Integrates with GST Suvidha Provider (GSP) APIs to fetch live and historical GSTR-2A / 2B directly from the GSTN portal using secure OAuth tokens.

Layer B — The Smart-Matching Engine

- Normalises messy text first: strips special characters, removes leading zeros, standardises state codes and legal names.
- Runs a multi-pass comparison chain across hundreds of thousands of line items and sorts every invoice into one of four buckets (see Section 9).

Layer C — Automated Communication & Recourse

- Groups all "missing in portal" and "mismatched" invoices by the responsible supplier.
- Auto-drafts and dispatches contextual messages by Email, WhatsApp and SMS, attaching a clean statement of the exact discrepancy, plus a self-service micro-portal where the vendor uploads or corrects the data.

Layer D — Executive Compliance & Financial Dashboard

- Shows, in real time, how much capital is locked up by vendor defaults — the "Unclaimed ITC Pipeline."
- Generates audit-ready reconciliation sheets in GSTR-9 / 9C layouts, exportable in one click for auditors or the department.

9. The Matching Pipeline, Step by Step

This is the core algorithm. It is intentionally ordered from strictest to loosest so that confident matches are made deterministically and only the leftovers reach the fuzzy logic.

1. **Pass 1 — Exact match.** Compare GSTIN + invoice number + date + tax heads. Perfect agreement → Fully Matched.
2. **Pass 2 — Normalised match.** Re-run on cleaned strings (delimiters/leading zeros stripped, state codes and names standardised) to catch INV/26-27/042 vs. 26-27-42.
3. **Pass 3 — Financial tolerance.** Allow small, rule-bound differences (e.g. rounding) on tax/taxable value, recording the net impact rather than failing the row.
4. **Pass 4 — Fuzzy / proximity.** For whatever is still unmatched, use proximity text logic to surface Probable Matches with a confidence score for human review.

Whatever survives all four passes is bucketed as follows:

Bucket	Meaning	Action
Fully Matched	All fields agree.	Cleared for GSTR-3B.
Mismatched	Found, but tax / value / GSTIN differs.	Flag net impact; notify vendor.
Missing in Portal	Buyer has the invoice; supplier never reported it.	Defer credit; chase vendor.
Missing in Books	Supplier reported it; buyer has no such entry.	Investigate possible error / wrongful filing.

10. Target Performance & Metrics

Metric	Target
Scalability	500,000+ purchase rows reconciled in under 2 minutes for large multi-location GSTINs.
Matching accuracy	99.8% precision using deterministic + financial-tolerance + proximity logic.
Financial yield	Typically recovers 1.5%–4% of total annual indirect-tax spend otherwise lost to vendor non-compliance.

Part III — The Commercial Pitch

11. The Elevator Pitch

> *For enterprises losing crores in unrecovered Input Tax Credit due to non-compliant suppliers, our platform is an automated Financial Recovery Engine that bridges the gap between your accounting books and the GST portal. Unlike traditional filing software that simply reports errors on a spreadsheet, it uses an intelligent multi-pass matching engine to instantly capture leaking tax credits, tracks ageing claims to prevent statutory lapses, and runs automated legal follow-ups with defaulting vendors until your money is recovered.*

12. The Legacy Blind Spots

Every enterprise on the legacy setup faces the same three operational blind spots — and each maps directly to a feature we built.

Legacy blind spot	The problem	Our answer
The spreadsheet trap	Manual v-lookups across millions of rows cause fatigue and hidden errors.	Background engine processes 100k+ rows in minutes.
The communication break	Fragmented emails and calls to vendors mean broken threads and unrecovered credit.	Automated multi-channel escalation loop + vendor portal.
Rigid static logic	Exact-only matching cannot handle invoice-string variation without human eyes.	Multi-pass: deterministic + tolerance + fuzzy text.

13. High-Value Commercial Pillars — the “Why Buy”

I. Immediate ROI — leaked-capital reclamation

The platform treats compliance as a revenue-recovery stream, not a cost centre. By locating missing invoices and matching obscured string patterns that legacy systems drop into the “unmatched” bucket, it directly recaptures 1.5%–4% of an enterprise's total annual indirect-tax spend — straight to net profitability.

II. Autonomous vendor recourse — zero manual chasing

Instead of manual emails, the platform converts the mismatch report into action. It provisions an outbound notification queue (Email, WhatsApp, SMS) that alerts the supplier's finance desk to the precise erroneous invoice, and provides a self-service micro-portal for them to upload or correct the data — effectively outsourcing the fix back to the party that caused it.

III. Pre-emptive audit defence

The software acts as a continuous internal auditor. It monitors the absolute deadlines under Section 16(4), flags ageing unclaimed credits before they lapse, and maintains an immutable, granular change log of every force-match and adjustment — so when a scrutiny notice arrives, an audit-ready reconciliation statement is one click away.

14. Legacy vs. Our Platform — At a Glance

Dimension	Legacy compliance software	Our recovery platform
Primary focus	Filing returns on time.	Recovering lost tax capital & ensuring vendor compliance.
Matching logic	Exact fields only (strict).	Multi-pass: deterministic + financial tolerance + fuzzy text.
Vendor disputes	Manual export to Excel → manual emails.	Automated multi-channel escalation loop + vendor portal.
Statutory safety	Static warning logs.	Proactive time-decay tracking for Section 16(4) & Rule 37.
Audit readiness	Reconstruct the trail under pressure.	Tamper-evident log; one-click ASMT-10 response.

Maximise working capital. Minimise audit risk. Recover what's yours.